



Review Article

DFT in catalysis: Complex equations for practical computing applications in chemistry

Artur Brotons Rufes^a, Sergio Posada Pérez^{a,b,*}, Albert Poater^{a,*} ^a Institut de Química Computacional i Catàlisi and Departament de Química, Universitat de Girona, c/ Maria Aurèlia Capmany 69, 17003, Girona, Spain^b Department of General Chemistry (ALGC), Vrije Universiteit Brussel (VUB), Pleinlaan 2, 1050, Brussels, Belgium

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ABSTRACT

Density Functional Theory (DFT) has become the cornerstone of modern computational catalysis, providing a practical balance between accuracy and efficiency in describing molecular structure, bonding, and reactivity. This review presents a comprehensive overview of DFT methodology, from its quantum-mechanical foundations and basis-set construction to the hierarchy of exchange–correlation functionals defined by Jacob's ladder. We discuss how DFT enables mechanistic elucidation of homogeneous and heterogeneous catalytic processes, highlighting benchmark studies that compare functional performance across representative reactions and transition states. Key interpretative tools, such as bond order analysis (Mayer, Wiberg, AIM/QTAIM), Natural Bond Orbital (NBO) theory, Energy Decomposition Analysis (EDA), and Non-Covalent Interaction (NCI) plots, are introduced as essential descriptors linking electronic structure to reactivity. The review also explores the integration of DFT with machine learning, microkinetic modeling, and automated reaction discovery, outlining recent advances toward predictive catalysis. Collectively, this work provides both conceptual and practical guidance for applying DFT to catalytic problems, emphasizing methodological awareness, descriptor-based interpretation, and emerging data-driven strategies for rational catalyst design. However, the main take-home message is that for DFT calculations, while in-depth methodological expertise is not essential, a clear comprehension of the theory's practical application is crucial.

1. Introduction

Where does chemistry begin? Answering this question requires identifying the simplest entity or phenomenon that can be studied within the framework of this discipline (Duncan et al., 2019). One might argue that the atom is the starting point, and this is accurate, as the periodic table remains the most iconic representation of chemistry in the public imagination. However, it may be even more precise to define chemistry as the behavior exhibited by different atoms when they interact. There is no chemistry in a static collection of atoms that do not engage in any interactions. The essence of chemistry lies in these interactions between atoms, which result in the formation and dissociation of molecules. This process can be encapsulated in a single concept: the chemical bond.

The chemical bond, with its various forms, is intrinsically dependent on the existence of the electron. Our understanding of the electron's nature is the result of numerous groundbreaking contributions,

primarily made during the first half of the 20th century (Fig. 1). Motivated by increasing discrepancies between classical models and experimental observations, quantum mechanics was developed. The formulation of the Schrödinger equation enabled the evaluation of this novel wave-like behavior of particles, necessitating a statistical framework to describe it accurately.

2. Schrödinger equation

The quantum-mechanical description of matter begins with the Schrödinger equation, which relates the total energy of a system to its electronic wavefunction ψ through the Hamiltonian operator $\hat{H}$:

$$\hat{H}\psi = E\psi \quad (1)$$

Eq. (1) contains, in principle, all the information needed to describe atoms and molecules. Analytical solutions exist only for the simplest

* Corresponding authors at: Institut de Química Computacional i Catàlisi and Departament de Química, Universitat de Girona, c/ Maria Aurèlia Capmany 69, 17003, Girona, Spain.

E-mail addresses: artur.brotons@protonmail.com (A. Brotons Rufes), sergio.posada.perez@vub.be (S. Posada Pérez), albert.poater@udg.edu (A. Poater).

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case, i.e. the H atom, because the explicit treatment of electron–electron repulsion in many-electron systems is mathematically intractable. Therefore, approximate approaches are required (Cramer, 2004).

A common simplification is to treat each electron as moving in the average field created by all the others while the nuclei remain fixed under the Born–Oppenheimer approximation. This mean-field concept leads to the Hartree and Hartree–Fock methods, where the total wavefunction is built from one-electron orbitals that provide a chemically intuitive picture of bonding.

Although the exact solution of the Schrödinger equation is unattainable for real molecular systems, its conceptual framework underpins every modern quantum-chemical method. In catalysis, these approximations make it possible to describe active sites, intermediates, and transition states at the electronic level, providing a direct connection between fundamental quantum theory and observed chemical reactivity.

2.1. Wavefunction

Most electronic structure methods approximate real systems by treating electrons as independent particles and then correcting for the missing interactions. This simplification reduces the many-body problem to a manageable pseudo-one-particle model. In this framework, one-electron functions, or orbitals, are introduced and used to construct approximate molecular orbitals (MOs) arranged in a Slater determinant. Since the hydrogen-like atom provides exact solutions to the Schrödinger equation, its orbitals serve as convenient one-electron functions to approximate atomic and molecular systems. In molecular calculations, the electronic wavefunctions are generally expressed as a linear combination of such atomic orbitals, an approach known as the Linear Combination of Atomic Orbitals (LCAO) scheme, where each molecular orbital is expanded as a weighted sum of atomic orbitals. The coefficients assigned to each atomic orbital indicate its contribution to the molecular orbital. This approach, although approximate, provides a practical and mathematically tractable foundation for understanding and computing the electronic structure of atoms and molecules, forming the conceptual basis of modern quantum chemical calculations.

2.2. The nature of the basis sets

Slater-type orbitals (STOs) were first introduced to model single-electron atomic functions, closely resembling hydrogen-like atomic orbitals through an exponential radial decay and hydrogenic angular dependence. These orbitals correctly reproduce the nuclear cusp and realistic electron density distribution. However, despite their physical accuracy, STOs present a major computational challenge: the four-index integrals required for molecular calculations lack analytical solutions, demanding numerical methods that are both time-consuming and prone to error (Mulliken et al., 1949; Bishop et al., 1966). To overcome this limitation, Gaussian-type orbitals (GTOs) were developed as an alternative. GTOs decay as a Gaussian function, allowing all necessary integrals to be evaluated analytically, thus drastically improving computational efficiency. Yet, their faster decay and lack of a cusp at the nucleus make them less realistic representations of true atomic orbitals. To address this, modern quantum chemical methods approximate STOs by combining multiple GTOs, forming contracted Gaussian functions that retain the computational advantages of GTOs while emulating the physical accuracy of STOs. This composite approach provides an optimal balance between precision and efficiency, serving as the foundation of nearly all contemporary basis sets used in molecular electronic structure calculations, including those implemented in software such as Gaussian and Amsterdam Density Functional (ADF) package (Te Velde et al., 2001; Rüger et al., 2023; Van Lenthe and Baerends, 2003).

2.3. Size of the basis set

From a mathematical perspective, increasing the size of the basis set improves the expressed function by increasing its flexibility. This is known as the multiple-zeta basis set, which allows a better description of bonding interactions with different symmetries. In organometallic DFT optimization, at least a double-zeta basis set is required for geometry optimizations. To make this more efficient, the additional basis functions are applied only to the valence orbitals, as they primarily describe chemical behavior. This results in a split-valence basis, focusing computational resources on the electrons involved in reactivity.

Furthermore, in cases where atomic orbital hybridization occurs (Fig. 2), additional considerations are required. A prime example is the ammonia molecule. Without the correct hybridization in the basis set, a

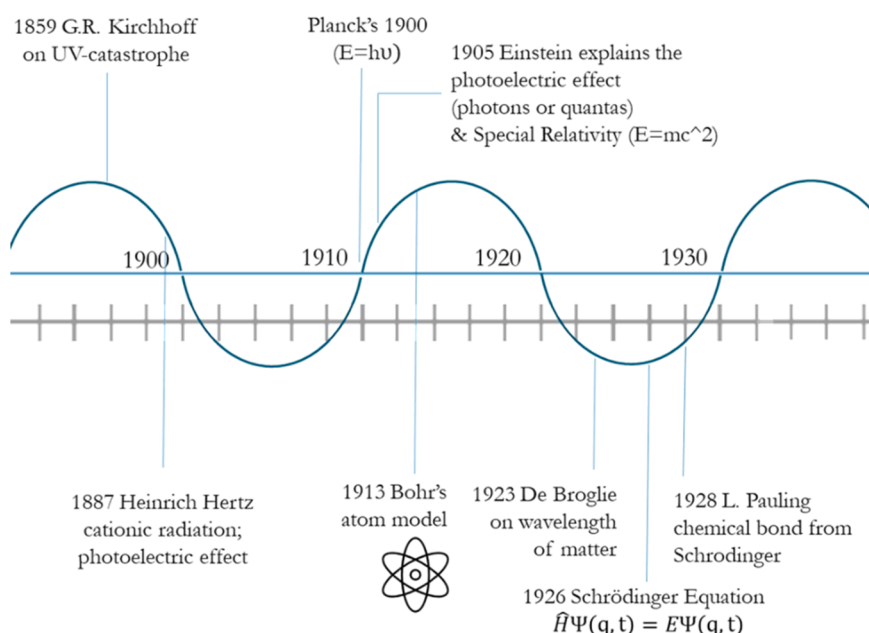


Fig. 1. Chronology of significant contributions to the Schrödinger equation (Duncan et al., 2019).

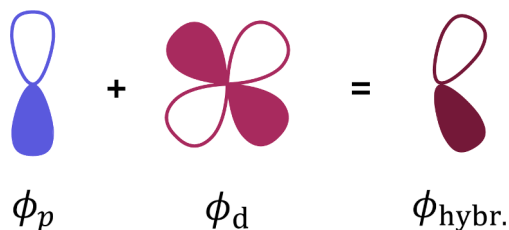


Fig. 2. The effect of adding higher-angular-momentum functions on orbital shapes.

HF optimization would predict a planar triangular structure (D3h) instead of the actual pyramidal shape. This discrepancy occurs because the nitrogen's real atomic orbitals cannot represent the hybridization in the molecule. To fix this, d-orbitals are added to nitrogen's basis set (Sun et al., 2013), which are referred to as polarization functions. These functions are generally added to all atoms in the system, except hydrogen. In electron-correlation methods, a more detailed polarization representation is required, which may involve higher-order polarization functions (Jensen, 2017).

An alternative method for incorporating polarization involves allowing basis functions to be located away from atomic centers. While this method avoids handling integrals related to high-angular-momentum functions, it significantly slows down geometry optimization. As a result, this method is seldom used in contemporary computations (Sun et al., 2013).

To avoid significant errors in energy calculations and other molecular properties, standard basis sets are often augmented with diffuse basis functions. This is especially important when modeling anions, highly excited electronic states, and loosely bound complexes. Without this flexibility, weakly bound electrons may not be able to localize properly, leading to inaccuracies (Sun et al., 2013).

When considering the antisymmetry of the electron's wavefunction using the Slater Determinant, the orbital is introduced as a factorization of the many-electron wavefunction into single-electron wavefunctions. In the hydrogenic atom, these orbitals emerge naturally since each energy level can only be occupied by one electron. Furthermore, according to Pauli's Exclusion Principle, each orbital can be occupied by a maximum of two electrons, provided they have opposite spins. This concept was introduced by Slater's determinant in a convenient form. However, neither this nor the determinant offers insight into how to select the appropriate orbital function. In the hydrogenic system, the emergence of AOs is a consequence of the underlying physics, making them a reasonable first approximation. This approach, known as LCAO, is presented above. Before proceeding further, the basis set section concludes with a method for describing core electrons in heavier atoms.

When introducing the split-valence concept, it was highlighted that core electrons typically have limited relevance in chemical descriptions. Similarly, pseudopotentials are used for heavier atoms, which contain numerous inner electronic shells. Explicitly including these shells in the wavefunction would lead to excessive computational costs with minimal improvement in model accuracy (Schwerdtfeger, 2011). Moreover, pseudopotentials provide an efficient way to account for relativistic effects in the core electrons of heavy elements (Dolg and Cao, 2012).

To clarify the terminology, pseudopotentials refer to a general method for approximating electron motion. When specifically applied to the inner shells of heavier elements, the term Effective Core Potentials (ECP) is used. ECPs simplify calculations by modifying the Hamiltonian operator, thereby improving computational efficiency. These modifications can also incorporate relativistic effects, which are critical for accurately modeling transition metals and heavy atoms. The remaining electrons are described using basis sets, which must be carefully optimized for compatibility with the chosen ECP (Fig. 3) (Kahn and Goddard, 1972, 1968).

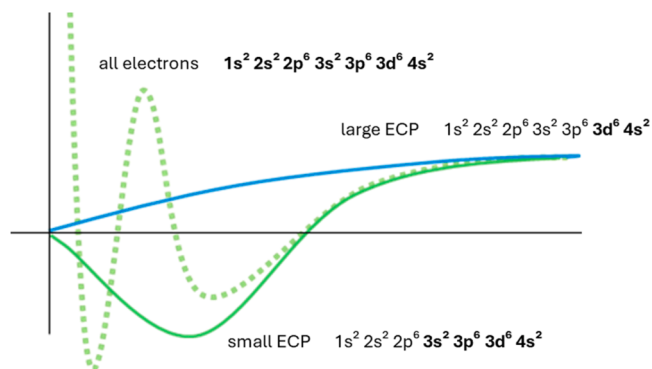


Fig. 3. Schematic representation of various pseudopotentials relative to the atomic radius.

3. Hartree Fock

The Hartree-Fock (HF) method provides the first practical approach to solving the many-electron Schrödinger equation. In HF theory, each electron is described as moving in the average electrostatic field created by all the others. The total wavefunction is represented as a single Slater determinant that automatically satisfies the Pauli exclusion principle.

The HF procedure is solved iteratively until self-consistency between orbitals and potential is achieved, yielding the so-called self-consistent field (SCF). This approach gives a mean-field description of the system, accurate enough to reproduce qualitative features of bonding and molecular geometry, but it neglects the instantaneous correlation between electrons.

Because electron correlation is not included, HF energies are systematically higher than experimental values. In practice, methods that correct for correlation, such as post-HF approaches or Density Functional Theory (DFT), are required to reach quantitative accuracy. DFT, in particular, provides a computationally efficient way to incorporate correlation effects, as discussed in the next section.

4. DFT

There are several reasons for the widespread use and growing popularity of DFT methods. Perhaps the most compelling is the physical nature of the electron density, which provides a direct connection to a real-world measurable quantity, in contrast to the more abstract concept of the wavefunction. In fact, the wavefunction is sometimes referred to as an oracle. When queried with an operator, it provides answers (Sun et al., 2013).

The Hartree-Fock method minimizes variationally the energy of a Slater determinant, which depends on the coordinates of all electrons, including three spatial and one spin coordinate per electron. This results in a large number of variables, particularly for systems with many electrons. By contrast, early DFT methods simplify the many-body problem by expressing it in terms of electron density, which depends solely on three spatial coordinates, irrespective of the number of electrons. As we shall see later, the introduction of Kohn-Sham orbitals reintroduces some dependence on the number of electrons.

4.1. DFT methodology and the Hohenberg-Kohn theorems

DFT is based on two foundational theorems formulated by Hohenberg and Kohn (Hohenberg and Kohn, 1964). The first theorem establishes that the electron density uniquely determines all properties of a system, including its external potential, Hamiltonian, and total energy. This means that for a given ground-state density, there can only be one corresponding potential and, therefore, one true energy. The proof relies on a reductio ad absurdum argument: assuming two different potentials could yield the same density leads to a contradiction. As a result, the

energy of a system can be expressed as a functional of its electron density, $E[\rho]$. The second theorem introduces a variational principle for the density. It states that any trial density corresponding to the correct number of electrons provides an energy value equal to or greater than the true ground-state energy. Consequently, by systematically refining the trial density, one could, in principle, approach the exact energy. In practice, however, the method faces two major challenges: there is no clear way to choose an accurate trial density, and calculating the energy as an expectation value reintroduces the complexity that DFT aims to avoid—namely, solving the many-body Schrödinger equation directly.

4.2. Kohn-Sham theory

Once it is established that electron density can serve as a variational parameter, the next step is to define the total energy as a functional of that density. The main challenge for non-hydrogenic systems lies in accurately describing electron–electron repulsion and quantum correlations. Kohn and Sham solved this problem by introducing a system of fictitious, non-interacting electrons that reproduce the same density as the real, interacting system (Lipkowitz and Boyd, 1996). This allowed them to separate the total energy functional into four components: the kinetic energy of non-interacting electrons, the external potential, the classical Coulomb (Hartree) term, and the exchange–correlation functional, which corrects for all remaining quantum effects. The exchange–correlation term accounts for both the difference between the true and approximate kinetic energy and the non-classical parts of electron–electron interactions. The Kohn–Sham formalism therefore simplifies the many-body problem into an effective one-electron framework that can be solved self-consistently using orbitals built from a Slater determinant. Despite its efficiency and success, the approach faces two main limitations: self-interaction errors must be corrected manually, and the exact analytical form of the exchange–correlation functional is unknown. This uncertainty drives the continuous development of approximate density functionals, which remain the central focus of modern DFT research.

4.3. DFT functionals and Jacob's ladder

Jacob's Ladder of DFT (Perdew and Schmidt, 2001), as defined by Perdew and collaborators, provides a hierarchical framework where the most accurate and complex Density Functional Approximations (DFAs) occupy the upper steps. The approximations are classified based on how the exchange–correlation functional $E_{xc}[\rho]$ is calculated. The topmost rung represents the Schrödinger equation for a fully interacting system (i.e., Full Configuration Interaction or FCI), solved without approximations. This framework suggests a positive correlation between the accuracy and complexity of DFAs, as supported by numerous benchmark studies (Mardirossian and Head-Gordon, 2017; Tran et al., 2016).

The first four rungs of the ladder use DFAs constructed from occupied orbitals:

- Local Density Approximation (LDA): Assumes the electron density is locally uniform, representing it as a homogeneous electron gas with slow density variations.
- Generalized Gradient Approximation (GGA): Builds on LDA by incorporating the local density gradient. While GGA provides additional physical insights, it does not always outperform LDA.
- Meta-GGA Functionals: Includes the Laplacian of the density, offering a more refined treatment of density variations.
- Hyper-GGAs (Hybrid Functionals): Combines a portion of nonlocal exact exchange energy with GGA, introducing nonlocality into the exchange–correlation functional. While hybrid functionals significantly enhance accuracy, their incorporation of exact exchange increases computational demands, especially in plane-wave calculations. Moreover, in Jacob's ladder of density functional approximations, double-hybrid functionals represent the fifth rung

(Perdew and Schmidt, 2001), combining conventional hybrid exchange–correlation with a perturbative correlation term, typically from MP2 theory. They provide improved accuracy for thermochemistry and noncovalent interactions.

It is noteworthy that the uniform electron gas remains the only scenario where the exchange–correlation functional $E_{xc}[\rho]$ can be derived exactly, given that the electron density is constant across all spatial points. More recently those functionals have been combined with dispersion corrections by Grimme and coworkers, by means of the D3 model (Grimme et al., 2010), and additionally with the Becke–Johnson (BJ) damping (Grimme et al., 2011), and followed by the Grimme's D4 dispersion correction, the successor to D3, which accounts for atomic charge dependence and improved environment sensitivity (Grimme et al., 2016; Caldeweyher et al., 2019). On the other hand, there are other approaches reported in the literature that also improve the description of dispersion interactions, such as the nonlocal correlation functional VV10 (Sun et al., 2013), which is included in several functionals developed by the Head-Gordon group (ω B97M-V or ω B97X-V) (Mardirossian and Head-Gordon, 2016, 2014).

4.4. State of the art in DFT modeling

Modern hybrid and meta-hybrid functionals, such as B3LYP (Becke, 1993; Lee et al., 1988; Stephens et al., 1994), PBEO (Adamo and Barone, 1999; Perdew et al., 1996; Ernzerhof and Scuseria, 1999), M06 (Zhao and Truhlar, 2008b, 2005, 2008a), and ω B97X-D (Mardirossian and Head-Gordon, 2016; Chai and Head-Gordon, 2008; Lin et al., 2013), balance computational cost and accuracy for a wide range of systems. These functionals have proven particularly effective for organometallic chemistry and catalysis, where electron correlation, spin states, and relativistic effects are crucial.

The evolution of DFT in catalytic modeling has been shaped by the growing capacity to handle complex reaction networks and large catalytic ensembles. Early DFT studies focused on simplified models with truncated ligands and idealized conditions (Ziegler and Rauk, 1977; Ziegler, 1991). Today, advanced functionals combined with dispersion corrections (e.g., D3(BJ), D4) (Grimme et al., 2010, 2011, 2016; Caldeweyher et al., 2019) and implicit solvation models (SMD (Marenich et al., 2009), PCM (Tomasi and Persico, 1994; Barone and Cossi, 1998)) allow accurate treatment of realistic catalytic systems involving large ligands, explicit solvents, and dynamic pathways.

Furthermore, recent progress in relativistic DFT, especially using effective core potentials (ECPs) (Hay and Wadt, 1985; Wadt and Hay, 1985; Andrae et al., 1990) and scalar relativistic corrections (ZORA, DKH) (van Lenthe et al., 1993; Douglas and Kroll, 1974; Hess, 1986), has improved the accuracy of predictions for heavy transition metals such as Pd, Pt, Au, and Ir, crucial for understanding complex processes.

Among the most recent insights on DFT, we highlight seminal studies that demonstrate its central role in unveiling catalytic mechanisms across organometallic, enzymatic, and heterogeneous systems. Data-driven frameworks such as the work by Mok et al. exemplify the integration of DFT and machine learning for CO₂ reduction electrocatalysis (Hyeon Mok et al., 2023), while Butera provided a comprehensive review bridging theory and practical catalysis (Butera, 2024). Shambhawi et al. explored computational design strategies for heterogeneous catalysts (Shambhawi et al., 2024), and Zhang et al. analyzed Fe- and Cu-based single-atom catalysts for the reverse water–gas shift reaction (Zhang et al., 2023). In parallel, Chen summarized DFT-based approaches for CO₂ conversion on MOFs (Chen, 2022), while Zhang L. et al. coupled experiment and DFT to rationalize Ni-doped Mo₂C electrocatalysts (Zhang et al., 2022). On the molecular side, Mavroskoufis et al. applied (TD-)DFT to photo-NHC catalysis (Mavroskoufis et al., 2023), and Nandy et al. used DFT to decode enzymatic blueprints for C–H activation (Nandy et al., 2022). Complementary studies by Hojaji et al. (2022) and Chowdhury et al. (2024) extended DFT applications to

energy storage and dual atom catalyst design. Collectively, these works underscore the versatility of modern DFT, from mechanistic elucidation to inverse catalyst design, showcasing its predictive and unifying role across the full spectrum of catalysis research.

Locating and verifying transition states (TSs) is central to mechanistic elucidation of DFT studies. Early strategies relied on chemical intuition to guess plausible TS geometries, followed by energy scans or constrained optimizations. Modern algorithms, however, have automated this process. Among the most widely used are the Nudged Elastic Band (NEB) method (Henkelman et al., 2000), the String Method (Sheppard et al., 2008), and the Growing String Method (GSM) (Jafari and Zimmerman, 2017), which map the minimum energy path between reactants and products. Combined with DFT gradients, these methods locate saddle points efficiently even for complex, multi-step reactions.

Vibrational frequency analysis remains the standard criterion for confirming transition states, with one imaginary frequency corresponding to motion along the reaction coordinate. Intrinsic Reaction Coordinate (IRC) calculations further connect TS structures to adjacent minima, ensuring mechanistic continuity. Recent developments have focused on automated TS search algorithms, such as AutoTS, TSSFinder, and the GRRM (Global Reaction Route Mapping) program (Maeda et al., 2013), which integrate artificial intelligence and global optimization to explore reaction landscapes with minimal user input.

Machine-learning-accelerated TS prediction represents an emerging frontier. Neural networks trained on DFT data can predict approximate TS geometries and energy barriers, drastically reducing the need for costly quantum chemical optimizations. Frameworks such as SchNet, ANI-2x, and DeepMind's GNN-based approaches are capable of identifying reactive coordinates and estimating activation energies across chemical spaces with near-DFT accuracy (Jackson et al., 2021; Kim et al., 2024).

While individual transition states provide mechanistic snapshots, catalytic systems often proceed through multiple interconnected pathways. Mapping these networks requires systematic exploration of all possible intermediates and transition states, followed by kinetic analysis. Traditional reaction discovery approaches relied on manually constructed mechanisms, but modern methods automate this process through reaction network generation and machine-learning-aided exploration.

Automated reaction discovery tools, such as AutoMeKin, KinBot, and ChemDyME use algorithms to generate and evaluate potential reaction steps (Van de Vijver and Zádor, 2020). They apply energy-based pruning criteria to identify kinetically relevant pathways. Once the potential energy surface (PES) is mapped, microkinetic modeling integrates thermodynamic and kinetic data into rate equations, predicting overall turnover frequencies (TOFs) and selectivity trends.

This approach has been instrumental in heterogeneous catalysis, where DFT-based microkinetic models explain trends across metal surfaces and ligand environments. For example, Sabatier analysis and scaling relationships derived from DFT data enable volcano plots that correlate catalytic activity with adsorption energies (Quaino et al., 2014). These descriptors have guided the rational design of catalysts for hydrogen evolution (Feidenhans'l et al., 2024), CO₂ reduction, and ammonia synthesis.

The integration of DFT data with kinetic Monte Carlo (kMC) simulations (Stamatakis and Vlachos, 2011) and ab initio molecular dynamics (AIMD) has further enhanced the realism of reaction network modeling. AIMD captures finite-temperature and solvent effects, while kMC bridges timescales between molecular events and macroscopic kinetics (Marx and Hutter, 2009). The synergy between static DFT and dynamic simulations thus provides a comprehensive view of catalytic performance under realistic conditions.

DFT plays a dual role in catalyst design: it rationalizes experimental results and predicts new candidates with improved properties. In homogeneous catalysis, particularly with transition-metal complexes, DFT elucidates how ligand structure, electronic effects, and coordination

geometry influence each step of the catalytic cycle. By decomposing total energy barriers into electronic and steric contributions, DFT analyses of oxidative addition, transmetalation, and reductive elimination have revealed that electron-rich ligands facilitate oxidative addition (Hartwig, 2010), while sterically demanding ligands promote reductive elimination by stabilizing low-coordinate intermediates (Harvey et al., 2003). Descriptor-based approaches, such as the Tolman cone angle (Tolman, 1977) and percent buried volume (%V_{Bur}) (Cavallo et al., 2005; Poater et al., 2009) and the derivative steric maps (Falivene et al., 2016, 2019), now integrate DFT-computed parameters like frontier orbital energies and metal–ligand charge transfer to predict catalytic performance (Poater et al., 2013; Poater and Cavallo, 2009).

In heterogeneous catalysis, DFT has transformed the understanding of surface chemistry. The development of periodic DFT methods using plane-wave basis sets (as implemented in VASP (Kresse and Furthmüller, 1996), Quantum ESPRESSO (Giannozzi et al., 2009, 2017), or CASTEP (Clark et al., 2005)) allows accurate modeling of metal and oxide surfaces. Adsorption energies, diffusion barriers, and reaction mechanisms on catalytic surfaces can now be predicted with high precision. Furthermore, the advent of machine-learning potentials, such as Gaussian Approximation Potentials (GAP) (Bartók et al., 2010) and Neural Network Potentials (NNP) (Behler and Parrinello, 2007), extends DFT accuracy to molecular dynamics simulations involving thousands of atoms.

A major challenge in catalyst design is the vastness of chemical space. DFT-derived descriptors provide a pathway to efficient screening. Parameters such as d-band center (for surfaces) (Behler, 2016), HOMO–LUMO gap, charge transfer, and Fukui functions (for homogeneous catalysts) (Yang and Mortier, 1986) serve as quantitative predictors of activity and selectivity (Poater et al., 2009). Linear scaling relations and volcano-type correlations derived from these descriptors have been used to identify optimal catalytic materials across reaction families, for homogeneous and heterogeneous catalysis (Nørskov et al., 2009).

Recent developments integrate DFT with machine learning to accelerate catalyst discovery. Machine learning models trained on DFT-calculated descriptors predict adsorption energies, reaction barriers, or turnover frequencies at a fraction of the computational cost. For instance, the Open Catalyst Project (OCP) (Chanussot et al., 2021) and Materials Project databases (Jain et al., 2025) provide large-scale DFT datasets enabling deep-learning models to generalize across materials and reactions. Active learning frameworks iteratively retrain machine learning models based on DFT feedback, efficiently exploring chemical space while minimizing redundant computations (Ulissi et al., 2017).

Hybrid quantum–machine learning methods are emerging as the next frontier (Schütt et al., 2018), with machine learned quantum potentials bridging DFT and dynamics (Smith et al., 2018). Techniques like Δ -learning correct systematic DFT errors using data-driven corrections (Ramakrishnan et al., 2015), while surrogate machine learning potentials allow real-time exploration of potential energy surfaces (Behler, 2021). These approaches have reduced the computational barrier between DFT-level accuracy and large-scale kinetic simulations, enabling in silico design of catalytic cycles with near-experimental precision.

Despite remarkable progress, DFT faces inherent limitations in describing certain catalytic phenomena. Standard exchange–correlation functionals often struggle with self-interaction errors (Cohen et al., 2012), multi-reference character, and dispersion forces. Transition-metal complexes involving open-shell configurations or spin-crossing reactions remain particularly challenging (Harvey, 2014). The development of multi-reference DFT (Gagliardi et al., 2017), range-separated hybrids (Tawada et al., 2004), and double-hybrid functionals aims to overcome these deficiencies (Goerigk and Grimme, 2014).

The integration of DFT with experimental spectroscopies (XAS, IR, NMR) through computational spectroscopy and inverse design will further strengthen its predictive capability (Neese, 2012). The future of

catalyst design likely lies in automated DFT workflows coupled with machine learning (Jain et al., 2025), enabling closed-loop discovery platforms where theoretical predictions directly guide experimental synthesis and testing.

However, DFT also has some issues, and spin contamination represents one of the major limitations of standard DFT when dealing with open-shell or broken-symmetry metal complexes. In unrestricted DFT calculations, the spin eigenfunction is not pure, leading to an overestimation of spin polarization and an artificial mixing of spin states. This error is particularly critical in transition-metal catalysis, where multiple spin states often lie close in energy and influence reactivity and selectivity. A widely adopted correction scheme is the Yamaguchi spin-projection approach, which estimates the pure spin-state energy by extrapolating from contaminated unrestricted results (Yamaguchi et al., 1988). Alternative strategies include restricted open-shell DFT (RODFT) and broken-symmetry spin projection methods, which partially mitigate contamination but may still yield qualitative errors in systems with strong static (non-dynamical) correlation (Harvey, 2009). In such cases, typically involving diradicals, spin-crossing reactions, or multi-configurational transition states, single-determinant DFT becomes unreliable. To overcome these deficiencies, multireference approaches such as CASSCF (Complete Active Space Self-Consistent Field) or CASPT2 are recommended (Roos and Andersson, 1995), as they can explicitly account for near-degenerate electronic configurations. More recently, multireference DFT (MR-DFT) and DFT-in-DFT embedding schemes have been developed to combine DFT efficiency with multireference accuracy (Gagliardi et al., 2017), providing a balanced treatment of static and dynamic electron correlation in complex catalytic systems. On the other hand, fractional-occupation-density-based tool to visualize and quantify static electron correlation was proposed by Grimme and coworkers (Grimme and Hansen, 2015), enabling identification of by strongly correlated electrons and providing a single global metric for static electron correlation.

4.5. Comparative performance of DFT functionals for catalytic applications

In the field of computational catalysis, the choice of exchange-correlation functional within DFT is one of the most critical methodological decisions, as it directly affects geometries, adsorption energies, transition-state barriers, redox/hydride energetics and mechanistic ranking. Numerous benchmark studies now show that performance is highly context dependent: no single functional excels in all catalytic regimes (Butera, 2024). Below we summarize key findings from the literature and distill practical guidance for catalysis practitioners. For example, a study of transition-metal-oxygen bond dissociation across many metals found that GGA functionals like PBE or RPBE produced large errors (MAEs of 48–76 kJ/mol) whereas hybrid/meta-hybrid formalisms such as B2PLYP, TPSSH and B3P86 yielded MAEs < 30 kJ/mol (Iron and Janes, 2019). Another benchmark for adsorption on metal surfaces produced a database of experimental adsorption energies and compared several functionals including RPBE, BEEF-vdW, etc. (Wellendorff et al., 2012).

To sum up, for homogeneous organometallic systems (e.g., complexes of Pd, Ni, Rh with phosphines, N-heterocyclic carbenes or other ligands), the accuracy of barrier heights and thermochemistry strongly depends on the fraction of Hartree–Fock (HF) exchange, the inclusion of dispersion corrections, and the treatment of spin-state and multi-reference character. In a seminal benchmark of late-transition-metal reaction profiles (Pd + H₂, Pd + CH₄, Pd + C₂H₆) the authors found that so-called “kinetics” functionals (e.g., mPW1K, BMK) performed poorer for TM systems than general-purpose hybrids; instead, functionals such as PBE0, B1B95 and PW6B95 offered comparatively better performance (Quintal et al., 2006). A ligand-dissociation benchmark (WCCR10) similarly found PBE0 and TPSSH among the most accurate for coordination reactions, although individual errors still reached ~50 kJ

mol⁻¹ (Weymuth et al., 2014).

For redox/hydride chemistry, e.g., hydricities of 3d-metal hydrides, one study reported an MAE of ~1.4 kcal mol⁻¹ (~0.06 eV) when geometries were optimized at a GGA-D3 level and single-points were computed with PBE0-D3, implicitly suggesting PBE0-D3 is a strong candidate for hydride/ligand design chemistry (Boyчук and Wetmore, 2023; Morgante and Peverati, 2023). Another redox/acid–base benchmark found PBE0 yielded an average unsigned error (AUE) ~1.3 ± 1.2 kcal mol⁻¹, outperforming many other hybrid/meta-GGA methods in that domain (Zhao and Truhlar, 2008; Silva and Ramos, 2011). Hence, for homogeneous catalysis one may infer: hybrid or meta-hybrid functionals with moderate HF exchange (~20–30 %), plus dispersion corrections, deliver the best compromise between affordability and chemical accuracy in typical organometallic catalytic cycles. Caveats remain: open-shell or multi-reference cases may still challenge these methods, and basis/solvation/thermal corrections may dominate the remaining error budget.

In heterogeneous catalysis, modelling often involves periodic slabs of metal/oxide surfaces, adsorbate binding energies, and reaction barriers on surfaces. Key challenges include: proper description of metal–adsorbate bonding, van der Waals (vdW)/dispersion interactions, lateral adsorbate coupling, coverage effects and delocalized d-states.

In the well-known Wellendorff adsorption database for late transition-metal surfaces, no functional was found to be uniformly excellent; the authors emphasized ample room for improvements (Wellendorff et al., 2015). A more focused study on methane steam reforming species on Ni(111) compared plain GGA (PBE, RPBE) with dispersion-corrected GGA (PBE-D3, PBE-dDsC) and vdW-DF variants (optB86b-vdW, BEEF-vdW). The best results (lowest RMSD) came from the DFT-D variants, indicating that dispersion correction materially improves adsorption and reaction energetics on metal surfaces (Kirchhoff et al., 2021). A recent work on CO₂ electro-reduction on single-atom catalysts found that GGA/meta-GGA estimates (e.g., PBE) underestimated overpotentials versus hybrids such as PBE0; the hybrids gave improved quantitative agreement with high-level data (Yan and Nørskov, 2013). Thus, the pragmatic guidance for heterogeneous catalysis is: (i) use dispersion-corrected GGA (or vdW-aware) functionals for large periodic screening studies (cost-vs-accuracy trade-off), (ii) if computational resources permit, test hybrid functionals for key steps, i.e. rds, in order to validate the GGA result set, and (iii) always treat binding/adsorption energies with caution if they lie within ±0.2–0.4 eV of mechanistic cut-offs.

Catalytic mechanism modelling frequently hinges on computed barrier heights ($\Delta G^\ddagger$) and relative energetics of intermediates and transition states. The benchmark on activation energies of Ni/Pd-based catalysts (164 data points) found PBE0-D3 to offer the best performance (MAD ~1.1 kcal mol⁻¹) followed by PW6B95-D3 and B3LYP-D3 (~1.9 kcal mol⁻¹), with hybrid/meta-hybrid functionals outperforming plain GGAs and meta-GGAs (Goerigk and Grimme, 2011; Steinmetz and Grimme, 2013). Indeed, a recent tutorial review emphasized that choice of functional deeply affects mechanistic conclusions, and that sensitivity analysis across functionals is essential. Because mechanistic distinctions may depend on small energy differences (e.g., 0.1–0.3 eV ~2–7 kcal mol⁻¹), the error floor of a given functional becomes a key factor: if your predicted lowest energy barrier is only 3 kcal mol⁻¹ below its competitor but functional error margin is ±4 kcal mol⁻¹, mechanistic confidence is low. Accordingly, for reaction network modelling one should: (i) apply the same functional across all steps (to preserve error-cancellation), (ii) benchmark one or two key barriers with a higher-level functional or methodology, (iii) examine how mechanistic orderings shift with alternative functionals.

Modern catalyst design workflows increasingly leverage high-throughput DFT + descriptor screening + machine learning. In such contexts, computational cost is paramount, and functionals like r²SCAN-D4, B97-3c or low-cost GGA-D4 variants are gaining traction. A recent review noted that although hybrid/double-hybrid methods offer higher

accuracy, they remain impractical for large screening sets and hence one must judiciously trade off accuracy for throughput (Butera, 2024). Users should be mindful that descriptor errors propagate, and that the functional used to build the dataset ultimately influences downstream artificial intelligence model predictive reliability.

Several caveats must temper the above guidance. First, many benchmark sets reveal substantial scatter: for example, the WCCR10 ligand-dissociation dataset found deviations up to ~ 50 kJ mol⁻¹ even for the best functionals (Weymuth et al., 2014). Second, the best functional may depend on the type of chemical problem (adsorption vs bond cleavage vs redox) rather than simply on hybrid vs GGA classification. Third, environmental effects (solvation, temperature/entropy, coverage, support effects) may dominate error budgets and overshadow functional choice.

Looking forward, the development of non-empirical meta-GGAs (e.g., r²SCAN) and double-hybrid methods, combined with machine-learning correction schemes (Δ -learning) and rigorous uncertainty quantification (e.g., Bayesian ensemble functionals such as BEEF-vdW) offer a promising path toward more predictive catalysis modelling (Kirchhoff et al., 2021). Nevertheless, until a universal functional for catalysis emerges, practitioners must remain vigilant: perform sensitivity checks, document functional choice strategy, and clearly communicate methodological uncertainty.

To sum up, in Table 1, for homogeneous organometallic catalysis, where accurate reaction barriers and spin-state energetics are essential, hybrid functionals such as B3LYP, PBE0, and TPSSH are generally recommended. In all cases where ligands or adsorbates engage in significant noncovalent interactions, explicit dispersion corrections (D3 or D4) should be applied to ensure reliable energetics, even though geometrically the changes are not significant sometimes (Meconi et al., 2017). For large periodic heterogeneous systems, GGA functionals with dispersion (e.g., PBE-D3 or RPBE-D3) remain the pragmatic and widely adopted standard, balancing accuracy and computational feasibility.

The emerging class of meta-GGAs, particularly r²SCAN-D4, offers near-hybrid accuracy at GGA-level cost, making these functionals especially attractive for machine-learning-assisted high-throughput screening of catalytic materials. In contrast, double-hybrid functionals such as B2PLYP-D3 are best reserved for small benchmark models or energy refinements, given their significantly higher computational demands. Finally, the BEEF-vdW functional provides a distinctive advantage in quantifying uncertainty in catalytic screening by delivering statistical error estimates on adsorption energies and volcano-plot predictions, thereby improving the robustness of computational catalyst design.

5. Mechanistic descriptions

Quantum mechanical modeling has become an increasingly popular method for exploring the mechanisms of organometallic reactions. DFT calculations are commonly employed to identify intermediates and transition states along a reaction pathway. All potential reaction pathways are mapped within the multidimensional space known as the Potential Energy Surface (PES). This concept represents the total energy of a molecular system—including electronic and nuclear contributions—as a function of the atomic coordinates.

Within the PES framework, chemically significant structures such as reactants, intermediates, transition states, and products are identified as stationary points with zero-gradient. The relative Gibbs free energies of these points, under specified temperature and pressure conditions, define the reaction's energy landscape. The sequence of these energy values helps infer the reaction mechanism. Notably, the theoretical existence of the PES is a direct result of the Born-Oppenheimer approximation, which decouples the movement of electrons and nuclei.

Various factors can affect the accuracy of Gibbs energy profile calculations, which, in turn, influence the mechanistic interpretation of chemical reactions. While this manuscript does not aim to review

computational approaches in depth, it serves as a foundation for understanding the modeling methods chosen for the organo(metallic) systems studied in any catalytic study.

Some research groups aim to go beyond aiding the interpretation of experimental data by striving for predictive catalysis (Escayola et al., 2024; Monreal-Corona et al., 2023). The goal of predictive catalysis is to optimize catalytic processes by identifying and controlling key parameters that influence activity and selectivity, thereby enhancing experimental efficiency. DFT calculations are invaluable for elucidating catalytic mechanisms, guiding catalyst design, and optimizing yield and selectivity. While most advances in catalysis are achieved through time-intensive, costly, and environmentally damaging trial-and-error experiments, predictive catalysis offers a sustainable alternative. By providing molecular-level insights into the catalytic process and identifying key descriptors that govern activity and selectivity, this approach reduces experimental effort (Tomasini et al., 2022; Martínez-Fernández et al., 2025; Monreal-Corona et al., 2023; Joly et al., 2023). Correlations derived from molecular descriptors, combined with robust statistical analysis, can reveal the origins of selectivity and enable the generation of testable hypotheses (Monreal-Corona et al., 2023).

When investigating reaction mechanisms, it is essential to critically assess what constitutes evidence for a catalytic mechanism and how it is evaluated. Evidence supporting a particular mechanism does not guarantee the accuracy of a proposed model. This issue arises not only from the limitations of mechanistic evidence, such as kinetic or spectroscopic data, but also from the difficulty of imagining and evaluating all possible models. Take for instance, the predicted rate determining step or state (rds) could be verified to be assisted, leading to a decrease in the energy barrier. If it remains the rds, the discrepancy is only quantitative with respect to the predicted experimental temperature (Gimferrer et al., 2022). However, if the assistance changes the transition state defining the rds (Kozuch and Shaik, 2011), the reactivity would be fundamentally altered, and the resulting error would be both quantitative and qualitative. Such assistance may arise from a solvent molecule (Joly et al., 2023; Gimferrer et al., 2022; Luque-Urrutia and Poater, 2017; Hu and Brinck, 1999), a second catalytic moiety (Natongchai et al., 2022), even in a dual metal catalysis framework (Gómez-Suárez et al., 2016) or intramolecularly by a ligand (Poater et al., 2009), a base (Gu et al., 2021), the counterion, or any other species present in the reaction medium.

A proposed reaction mechanism should be regarded as a working theory, and its predictions must be validated against experimental results. Mechanisms must align with observable data, including overall reaction stoichiometry, selectivity, rate laws, activation parameters, intermediates, and their spectral signatures. Despite advances in technology that some suggest might eventually "prove" mechanisms, reinterpretation remains an ongoing possibility.

To advance the field of catalysis, researchers should focus on disproving rather than validating favored mechanistic theories. Mechanistic studies should aim to eliminate alternative explanations, thereby strengthening the credibility of hypotheses that remain unrefuted (Buskirk and Baradaran, 2009; Scott, 2019).

It is also important to recognize that computational methods do not inherently reveal reaction pathways. Instead, they evaluate reactions based on current chemical understanding (Dewyer and Zimmerman, 2017). Computational investigations often emphasize productive reaction pathways while overlooking factors like catalyst deactivation or off-cycle reactions (Balcells and Nova, 2018). This narrow focus means that only a fraction of the PES is explored. Additionally, multiple reaction paths can connect a specific reactant to a product, further complicating analysis.

5.1. Theoretical model

To determine equilibrium and rate constants, it is necessary to calculate Gibbs free energies, as these depend on Gibbs energy differ-

Table 1
Comparative performance of commonly used DFT functionals in catalytic modelling.

Functional	Type	Key Characteristics / Features	Principal Strengths	Principal Limitations	Recommended Catalytic Applications
PBE (Perdew–Burke–Ernzerhof) (1996)	GGA	Nonempirical; gradient-corrected exchange–correlation; standard in solid-state catalysis codes.	Robust and efficient; accurate structures; well-behaved for metallic systems; good for relative trends.	Lacks long-range dispersion; underestimates weak adsorption energies; poor barrier heights.	Metal/oxide surface catalysis; periodic slab calculations; geometry optimizations (Butera, 2024; Wellendorff et al., 2015).
RPBE (Revised PBE) (1999)	GGA variant	Modified exchange to reduce PBE overbinding; retains GGA efficiency.	Improved adsorption energetics for weakly bound species; widely benchmarked for surfaces.	Often over-corrects (underbinds strongly adsorbed species); lacks dispersion; poor for gas-phase kinetics.	Adsorption studies and volcano-plot screening on metal surfaces (Butera, 2024; Wellendorff et al., 2015).
B3LYP (Becke–Lee–Yang–Parr) (1993)	Hybrid (20 % HF)	Combines Becke 88 exchange + LYP correlation; semiempirical hybrid.	Excellent for main-group thermochemistry; widely validated for organic and organometallic reactions; good geometries and thermodynamics with D3 correction.	Lacks dispersion unless corrected; can mispredict spin states; not suitable for extended surfaces.	Homogeneous catalysis; ligand-field effects; redox and hydricity analysis (Zhao and Truhlar, 2008; Quintal et al., 2006).
PBE0 (PBE1PBE) (1999)	Hybrid (25 % HF)	Nonempirical hybrid of PBE exchange–correlation.	Good accuracy for reaction barriers, bond dissociation, and redox energies; balanced exchange fraction; excellent for single-point energies.	High cost for periodic models; overestimates barriers in correlated systems; requires dispersion correction.	Transition-metal complexes; oxidative addition/reductive elimination; homogeneous catalysis (Butera, 2024; Quintal et al., 2006; Weymuth et al., 2014).
TPSSH (2003)	Meta-hybrid (10 % HF)	Hybrid of TPSS meta-GGA with partial exact exchange; designed for TM systems.	Reliable for transition metals; improved spin-state energetics; accurate thermochemistry with low cost.	Slightly empirical; sometimes overstabilizes low-spin states; lacks explicit dispersion.	Organometallic reaction mechanisms; spin-crossover catalysis (Quintal et al., 2006; Weymuth et al., 2014).
M06-L (2006)	Meta-GGA (0 % HF)	Minnesota functional parameterized for kinetics and noncovalent interactions.	Reasonable for TM complexes; accounts for medium-range correlation; fast for geometry optimizations.	Overparameterized; less transferable; grid sensitivity; poor for metallic surfaces.	Screening of organometallic structures and intermediates; bulkier ligands (Zhao and Truhlar, 2008).
M06–2X (2006)	Meta-hybrid (54 % HF)	High exact-exchange fraction; calibrated for thermochemistry and kinetics.	Excellent for main-group barrier heights and dispersion; accurate for noncovalent interactions.	Unsuitable for metallic or delocalized systems; too much HF exchange for π -block metals.	Organic catalysis; nonmetal reaction mechanisms; small-molecule activation (Zhao and Truhlar, 2008; Silva and Ramos, 2011).
B2PLYP-D3 (2007)	Double hybrid	Combines DFT exchange with MP2-like correlation; dispersion-corrected variant available.	Very high accuracy for energies and barriers; systematic improvability.	Computationally demanding; poor for multi-reference systems; rarely implemented for periodic systems.	Benchmarking small homogeneous catalysts; validation of reaction profiles (Goerigk and Grimme, 2011).
r ² SCAN-D4 (2021)	Meta-GGA (nonempirical)	Regularized SCAN with improved numerical stability and dispersion (D4) correction.	Excellent geometry and thermochemistry accuracy; robust for both molecules and solids; near-hybrid quality at GGA cost.	Slightly less accurate for open-shell TM systems; newer, fewer catalysis benchmarks.	General-purpose functional for both homogeneous and heterogeneous catalysis; high-throughput workflows (Butera, 2024; Wellendorff et al., 2015).
BEEF-vdW (2012)	GGA + vdW + Bayesian ensemble	Parameterized for surface catalysis; includes vdW interactions and uncertainty estimation.	Good adsorption energies; built-in error estimation; ideal for scaling-relations and descriptor studies.	Poor for molecular TSs; expensive; may overbind weak adsorbates.	Heterogeneous catalysis on metal surfaces; uncertainty quantification (Butera, 2024; Wellendorff et al., 2015).

ences rather than internal energy differences. The relationship between internal energy (ΔE) and Gibbs energy (ΔG) is expressed as Eq. (2):

$$\Delta G = \Delta E_{\text{elec}} + \Delta H_{\text{therm}} - T\Delta S \quad (2)$$

Here, ΔE_{elec} refers to the electronic energy obtained from DFT calculations, with nuclear repulsion included. The term $(\Delta H_{\text{therm}} - T\Delta S)$ accounts for thermal and entropic contributions at a given temperature. Frequency calculations are required to determine the partition functions of the solute, providing the thermal and entropic corrections necessary to distinguish Gibbs energy from internal energy.

The accuracy of these calculations strongly depends on the choice of functional and basis set. Geometry optimization is a critical preliminary step, aiming to locate local minima for intermediates or local maxima for transition states. For systems with many atoms or transition metal centers, this process can be computationally expensive.

To enhance accuracy, it is common to refine electronic energy through single-point calculations after geometry optimization. This method isolates electronic energy calculations, offering a more efficient approach. The assumption that enthalpy and free energy differences remain consistent when using different functionals and basis sets further reinforces the reliability of results. The reaction free energy is ultimately computed as:

$$G \cong E_{\text{elec}}^{\text{BS2}} + (G^{\text{BS1}} - E_{\text{elec}}^{\text{BS1}}) \quad (3)$$

Solvent effects, which significantly influence the stability of chemical species, must also be incorporated into energy calculations. While explicitly including solvents in optimization is ideal, implicit solvent models are often employed in single-point calculations for efficiency. In detail in Eq. (3) there is the combination of the high-level electronic energy (computed with a large basis set, BS2) with thermochemical corrections (computed with a cheaper method/basis set, BS1) to approximate the final Gibbs free energy.

In most studies, entropy is often overestimated because calculations neglect the fact that molecules are not in a true liquid state in electronic terms (Rusconi et al., 2023). This discrepancy is typically corrected using implicit solvation models such as SMD (Marenich et al., 2009) or PCM (Tomasi and Persico, 1994; Barone and Cossi, 1998), which incorporate the solvent's radii and dielectric constant. On the other hand, while implicit solvation models offer approximate corrections, explicit solvent dynamics and entropic contributions require hybrid quantum/classical (QM/MM) (Senn and Thiel, 2009) or *ab initio* molecular dynamics simulations (Marx and Hutter, 2009). Additionally, real catalytic environments, such as electrochemical interfaces or operando conditions, demand inclusion of applied potentials and electric fields, a rapidly evolving area in computational electrochemistry (Chan and Nørskov, 2015; Gauthier et al., 2019).

In addition, a well-known limitation of standard DFT thermochemistry is the harmonic oscillator-rigid rotor approximation, which tends to overestimate entropic contributions, particularly for low-frequency modes associated with surface adsorbates or flexible molecular intermediates. To address this issue, Martin and co-workers proposed a quasi-harmonic correction scheme that replaces very low vibrational frequencies with an effective cutoff, thereby yielding more realistic entropy and free-energy estimates (Martin et al., 1998). Practically, they proposed a correction based on pressure, applying a maximum value of 1354 atm for water. This value is scaled according to the solvent's density using the ideal gas equation, incorporating the molecular weight of the solvent and temperature (Coufourier et al., 2020; Sadjadi et al., 2023), and it is important to compare dissociative with concerted mechanisms (Poater et al., 2014; Urbina-Blanco et al., 2013), in particular dual metal catalysis (Gómez-Suárez et al., 2016). Additionally, to account for the standard experimental concentration of 1 M, an extra correction term is introduced, which is temperature-dependent and was proposed by Cramer, Truhlar and coworkers (Chamberlin et al., 2006), and widely used since then (Acuña-Parés et al., 2014).

Among other approaches (Solans-Monfort et al., 2012; Falivene et al., 2018), subsequent studies have shown that such corrections can substantially modify the shape of reaction free-energy profiles and even alter the predicted rds. For example, in heterogeneous catalysis, corrected entropic contributions were found to lower apparent barriers and improve agreement with experimental turnover frequencies (González-Fabra et al., 2019). Similarly, recent computational studies demonstrated that applying quasi-harmonic or hindered-rotor corrections to entropies can reshape the Gibbs free-energy landscape in organometallic reaction mechanisms, leading to refined predictions of selectivity and kinetics (Ariai and Gellrich, 2023). Incorporating such physically motivated entropy corrections is therefore essential for accurate thermodynamic and kinetic modeling, particularly when comparing intermediates and transition states with distinct conformational flexibility or adsorption character.

5.2. Catalytic systems and energy profile interpretation

Interpreting energy profiles of reactions has traditionally been achieved through Eyring's Transition State Theory (TST), focusing on the rds (i.e., the step with the lowest energy barrier) (Boyd, 1978; Murdoch, 1981). However, the context of a stoichiometric reaction differs significantly from that of a catalytic process. Kozuch and Shaik have argued for incorporating the cyclic nature of catalytic processes into thermodynamic evaluations (see Fig. 4).

In their 2011 account, they highlight the limitations of the rds descriptor, which is often misleading (Laidler, 1988; Northrop, 1981). Under steady-state conditions, all steps in a catalytic cycle progress at equivalent rates by definition. Kozuch advocates for the Energy Span Model (ESM), which offers a more comprehensive evaluation of catalytic cycles (Kozuch and Shaik, 2011, 2008; Kozuch and Martin, 2011; Uhe et al., 2011). The rate of a single-step reaction can be expressed via the Arrhenius equation (Besalú i Llorà, 2004), with the energy span (δ_E) concept extending this to multi-step catalytic cycles. Unlike consecutive intermediate-transition state pairs, δ_E considers the largest energy gap in the cycle (see Fig. 4).

$$E_a + \Delta E = \delta E \quad (4)$$

The Turn-Over Frequency (TOF), a critical measure of catalytic efficiency, can also be derived theoretically using the ESM:

$$TOF_{\text{exp}} = \frac{TON}{\text{time}} = \frac{\text{mols}_{\text{product}}}{\text{mols}_{\text{catalyst}} \cdot \text{time}} \quad (5)$$

$$TOF = \frac{k_B T}{h} \frac{e^{-\Delta G_r} - 1}{\sum_{i,j=1}^N e^{(T_i - T_j - \delta G'_{ij})/RT}}, \text{ with } \delta G'_{ij} = \begin{cases} \Delta G_r & \text{if } i > j \\ 0 & \text{if } i \leq j \end{cases} \quad (6)$$

$$TOF = \frac{k_B T}{h} e^{-\delta E/RT} \quad (7)$$

$$\delta E = \begin{cases} TS_{\text{TSTS}} - I_{\text{TDI}} & \text{if TSTS appears after TDI} \\ TS_{\text{TSTS}} - I_{\text{TDI}} + \Delta G_r & \text{if TSTS appears before TDI} \end{cases} \quad (8)$$

Here, two key terms are introduced: the TOF-determining transition state (TSTS) and the TOF-determining intermediate (TDI). These correspond to the states that maximize the energy span (δE), serving as the apparent activation energy for the cycle. These terms correspond to the transition state and intermediate that define the maximum energetic span within the cyclic system. It is worth noting that Eq. (6) provides the exact expression, whereas Eq. (7) gives the approximation. Their primary role is to accurately describe the kinetics of the catalytic cycle. Additionally, δE , expressed in terms of Gibbs free energy, represents the apparent activation energy governing the overall turnover frequency. Consequently, a smaller energetic span correlates with a faster catalytic reaction. This kinetic-thermodynamic relationship has been widely applied to interpret reaction profiles across various organometallic

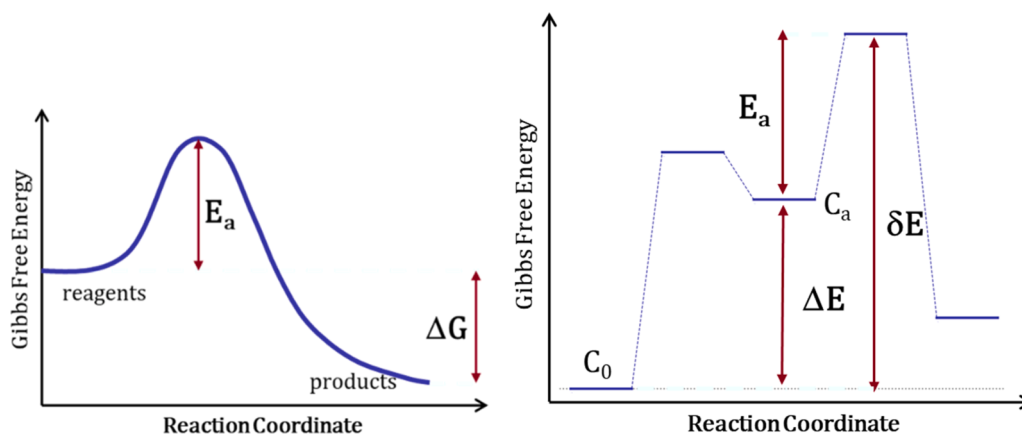


Fig. 4. Energy activation of a reaction based on the Arrhenius equation (E_a = activation energy) on the left; graphical representation of the estimated energetic span (δE), defined as the energy gap between the cycle's "peak and valley" (C_a and C_0 , respectively), on the right.

systems. For instance, Tobisch demonstrated the correlation between energetic span and turnover rate in zirconium-catalyzed amino-allene hydroaminations (Tobisch, 2008), while Normand and Hawkes explored related kinetic analyses in Ni-NHC and Rh-catalyzed C–C coupling reactions (Normand et al., 2007; Hawkes et al., 2008). Likewise, Liu and co-workers illustrated how substituent and ligand effects modulate the energetic span in Rh(I)-catalyzed (5 + 2) cycloadditions (Liu et al., 2008), and Guiducci et al. extended such concepts to CO₂ insertion reactions with Ti-imido complexes (Guiducci et al., 2009).

These equations challenge conventional interpretations of kinetics in several ways. As mentioned earlier, reaction kinetics are not solely determined by a single step (i.e., a consecutive intermediate and transition state) but rather by the states that define the largest energy gap (i.e., the energetic span). Therefore, rate determining steps do not exist—only rate determining states. Moreover, these rate determining states are not necessarily the highest transition state or the lowest intermediate, but rather those that maximize δE (see Fig. 5) (Murdoch, 1981; Ray, 1983). Finally, no single transition state dictates the cycle's kinetics; instead, the catalyst's influence must be considered across all intermediates involved. In 2015, building on earlier kinetic formulations, Kozuch introduced a graph-theoretical representation of ESM that enables automated identification of the TDTS and TDI in complex catalytic networks (Kozuch, 2015).

Illustrative real-world examples where DFT has provided crucial mechanistic and catalytic insights include (i) elucidation of C–H activation mechanisms in organometallic catalysis, (ii) identification of CO₂ reduction pathways on transition-metal and heterogeneous catalysts,

and (iii) prediction of enantioselectivity in asymmetric catalysis (Harvey et al., 2019; Lonsdale et al., 2012). These case studies highlight how DFT enables the identification of key intermediates and transition states, rationalizes selectivity trends, and supports the rational design of improved catalysts. For instance, DFT investigations on C–H activation by Cp*Ir complexes have identified key transition states and clarified the role of ligand electronics and sterics in controlling selectivity (Oxgaard et al., 2004). Similarly, Beller and co-workers employed DFT to uncover the mechanism of CO₂ hydrogenation to methanol over Cu–ZnO–Al₂O₃ catalysts, revealing the rds and rationalizing the superior performance of Cu surfaces (Behrens et al., 2012; Kattel et al., 2017). In the realm of homogeneous catalysis, Sigman and coworkers used DFT-supported multivariate models to achieve predictive enantioselectivity control in asymmetric C–H functionalization and cross-coupling reactions (Santiago et al., 2018). Furthermore, Harvey and Maseras demonstrated the use of the Energetic Span Model to correlate computed free-energy profiles with experimentally observed turnover frequencies (Harvey et al., 2019), providing a quantitative bridge between DFT data and catalytic performance. Collectively, these studies exemplify how DFT serves not only as an interpretative tool but also as a predictive framework for catalyst optimization and design.

6. Characterization techniques

6.1. Bond orders

Quantifying bond strength is essential for characterizing a chemical

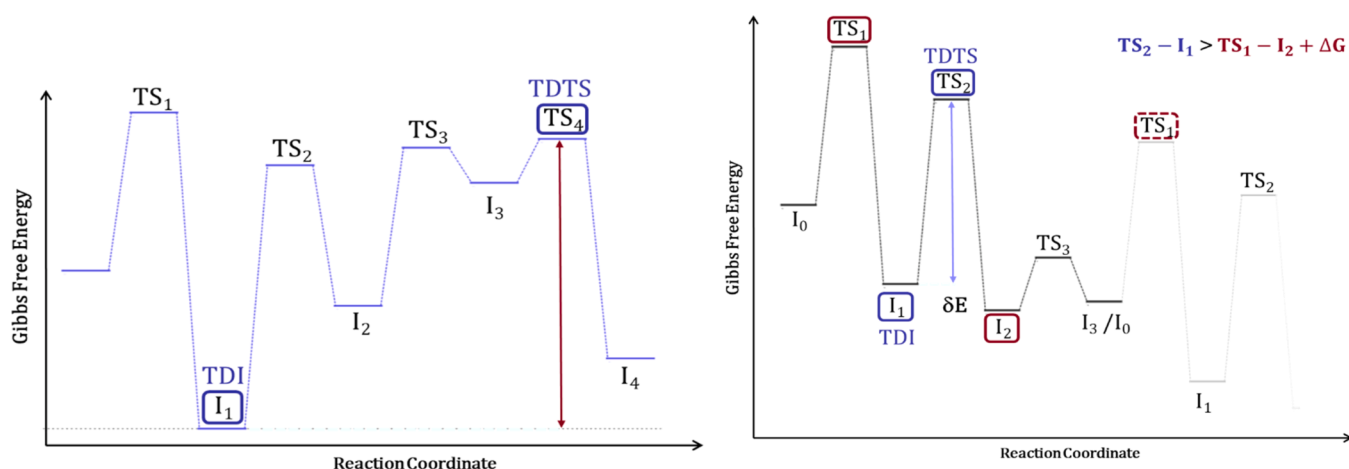


Fig. 5. Rate determining step where the rate determining transition state is after (left) or before (right) the rate determining intermediate.

system. While bond distance provides some insight, the complexity of chemical bonding necessitates more advanced methodologies (Gimferrer Andrés, 2023). Quantum mechanical approaches, such as HF and DFT, offer detailed information through the wave function, enabling a deeper understanding of the system. Methods that extract bond characteristics directly from the wave function (or its density) after optimizing HF or KS-DFT calculations are particularly valuable (Casademont-Reig et al., 2021).

At this stage, it is important to define key concepts: one-electron density and two-electron density, along with their corresponding reduced density matrices (Bader, 1991, 2003). One-electron density describes the probability of locating an electron at a specific point in space. It is mathematically derived from the system's wave function. For an (N)-electron system, the one-electron density ($\rho(r)$) is given by:

$$\rho(r_1) = N \int |\Psi(r_1, r_2, \dots, r_N)|^2 dr_2 \dots dr_N \quad (9)$$

where $\Psi(r_1, r_2, \dots, r_N)$ represents the many-electron wave function, and the integration is performed over all electron coordinates except r_1 . Because electrons are indistinguishable, the factor N accounts for all electrons in the system.

The two-electron density, or pair density, provides information about the probability of finding two electrons at specific locations simultaneously. It is defined as:

$$\rho_2(r_1, r_2) = N(N-1) \int |\Psi(r_1, r_2, r_3, \dots, r_N)|^2 dr_3 \dots dr_N \quad (10)$$

Here, normalization is applied using the McWeeny-Dirac factor $N(N-1)$. The two-electron density is crucial for understanding electron correlation effects, which significantly influence the accurate description of electronic behavior in a system.

Notably, higher-order n -electron density matrices are not required to describe electron-electron interactions, since the two-electron reduced density matrix (2-RDM) fully determines all pairwise correlation effects and, together with the one-electron density (Mazziotti, 2012), contains all the information needed to compute the total electronic energy (Gimferrer Andrés, 2023).

While the electron density, $\rho(r)$, provides a spatially resolved picture of where electrons are most likely to be found, the one-electron reduced density matrix (1-RDM), $\gamma(r, r')$, contains additional phase and off-diagonal information that reflects electronic coherence and delocalization between different regions of space (McWeeny, 1960; Löwdin, 1955). It is mathematically defined as:

$$\gamma_1(r_1, r_1') = N \int \Psi^*(r_1, r_2, \dots, r_N) \Psi(r_1', r_2, \dots, r_N) dr_2 \dots dr_N \quad (11)$$

Here, r_1 and r_1' represent different spatial coordinates of the same electron, making $\gamma_1(r_1, r_1')$ dependent on two sets of coordinates.

Similarly, the two-electron reduced density matrix is given by:

$$\gamma_2(r_1, r_2; r_1', r_2') = N(N-1) \int \Psi^*(r_1, r_2, r_3, \dots, r_N) \Psi(r_1', r_2', r_3, \dots, r_N) dr_3 \dots dr_N \quad (12)$$

For a single-determinant system (e.g., where the wave function is represented by a single Slater determinant, such as in HF or KS-DFT), this expression simplifies to:

$$\gamma_2(r_1, r_2; r_1', r_2') = \begin{vmatrix} \gamma_1(r_1, r_1') & \gamma_1(r_1, r_2') \\ \gamma_1(r_2, r_1') & \gamma_1(r_2, r_2') \end{vmatrix} \quad (13)$$

This determinant structure represents the probability amplitude of two electrons occupying specific positions, with γ_1 being the fundamental component.

These density matrices allow for a detailed analysis of a system's electronic structure and bonding characteristics. Importantly, these

concepts are not new, as they form the foundation for calculating electronic energy terms within the KS-DFT framework. Specifically, kinetic energy is a functional of the 1-RDM, electron-nucleus potential can be derived from electron density, and electron-electron repulsion is determined from pair density.

Defining a chemical bond requires considering the relative interactions of the two electrons involved. This information is contained within the pair density, but extracting it typically involves using an alternative pair function. In single determinant wavefunction, this function is often the exchange-correlation density (XCD) (Chen and Friesecke, 2015):

$$\rho_{xc}(r_1, r_2) = \rho(r_1)\rho(r_2) - \rho_2(r_1, r_2) \quad (14)$$

Here, $\rho(r_1)\rho(r_2)$ represents a fictitious pair density for a non-interacting system, meaning that $\rho_{xc}(r_1, r_2)$ captures only the strict correlation present in $\rho_2(r_1, r_2)$.

Since the exchange density depends on the coordinates of both electrons, the total electron count can be determined by integrating over all coordinates:

$$N = \int \rho_{xc}(r_1, r_2) dr_1 dr_2 \quad (15)$$

XCD plays a central role in various widely used bonding analysis tools, including electron-sharing indices, or bond orders (Ruedenberg, 1962), which quantify the number of chemical bonds between two atoms.

Using the linear combination of atomic orbitals (LCAO) formalism, the exchange density can be expressed in terms of molecular orbitals (MOs):

$$\rho_{xc}(r_1, r_2) = 2 \sum \phi_i^*(r_1) \phi_j(r_2) \phi_j(r_1) \phi_i(r_2) \quad (16)$$

where the molecular orbitals (ϕ_i) are given by:

$$\phi_i = \sum c_{\mu i} \chi_{\mu} \quad (17)$$

Here, $D_{\mu\nu}$ represents the density matrix in the atomic orbital (AO) basis, defined from the one-electron density:

$$\rho_1(r_1) = \sum D_{\mu\nu} \chi_{\nu}(r_1) \chi_{\mu}(r_2) \quad (18)$$

with the density matrix components expressed as:

$$D_{\mu\nu} = 2 \sum C_{\mu i} C_{\nu i} \quad (19)$$

For a closed-shell restricted system, this holds true. When considering alpha and beta electrons separately, the density matrix is divided into its alpha (D^{α}) and beta (D^{β}) components, with the total density matrix given by:

$$D_S = D^{\alpha} + D^{\beta} \quad (20)$$

The spin density matrix (D_S) is defined as:

$$D_S = D^{\alpha} - D^{\beta} \quad (21)$$

with individual spin components calculated as:

$$(D^{\alpha})_{\mu\nu} = \sum C_{\mu i} C_{\nu i}, i \in occ^{\alpha} \quad (22)$$

$$(D^{\beta})_{\mu\nu} = \sum C_{\mu i} C_{\nu i}, i \in occ^{\beta} \quad (23)$$

Additionally, the overlap matrix S is given by:

$$S = \int \chi_{\mu}(r_1) \chi_{\nu}(r_1) dr_1 \quad (24)$$

Mayer proposed decomposing the exchange density, similar to the Mulliken overlap population, into overlap and density matrix components (Mayer, 1985). Consequently, the normalization expression becomes:

$$N = \int \int \rho_{xc}(r_1, r_2) dr_1 dr_2 = \sum [(D_s)_{\mu\nu} (D_s)_{\nu\mu} + (D_s S)_{\mu\nu} (D_s S)_{\nu\mu}] \quad (25)$$

The Mayer bond order (MBO) is then defined as:

$$B_{AB} = \sum \sum [(D_s)_{\mu\nu} (D_s)_{\nu\mu} + (D_s S)_{\mu\nu} (D_s S)_{\nu\mu}] \quad (26)$$

where the summations apply only to the subspaces corresponding to atoms A and B.

For a closed-shell wavefunction, $D_s = 0$, causing the second term in the summation to vanish. However, the MBO method has certain limitations, such as strong basis set dependence and unrealistic values when using large basis sets with diffuse-type atomic orbitals. These inaccuracies stem from the shape of the atomic orbitals in the chosen basis set.

To mitigate this issue, the basis set can be transformed into a more suitable form using linear algebra. One approach is Löwdin symmetric orthogonalization, where the unitary matrix U is the inverse square root of the overlap matrix (Löwdin, 1950):

$$U = S^{1/2} \quad (27)$$

Using bracket notation for clarity:

$$|\varphi_{Low}\rangle = U|\chi\rangle = S^{1/2}|\chi\rangle \quad (28)$$

where $\delta_{\mu\nu}$ is the Kronecker delta.

Expressing bond order in this orthogonal basis leads to the Wiberg bond index (WBI) (Wiberg, 1968):

$$B_{AB} = \sum \sum D_{\mu\nu}^{Low} \quad (29)$$

The diagonalization of the basis set within the Löwdin framework makes it more suitable for atomic charge calculations as well. And thus, the WBI is often preferred due to its improved accuracy, sensitivity, and broader applicability in chemical systems (Outeiral et al., 2018). Nevertheless, both indices are often used interchangeably (Müller et al., 2021; Sizova et al., 2008). However, WBIs are more commonly employed to describe reactivity (Ugur et al., 2017; Stambuli et al., 2002), while MBOs particularly in applications in catalysis (Poater and Cavallo, 2010; Poater et al., 2011). One of the primary applications is the comparison of bonds involving atoms of different chemical nature (Poater et al., 2014; Hanifpour et al., 2020; Kaur et al., 2017; Poater et al., 2018), including material science (Ahmadi et al., 2022; Dehghani et al., 2019). In particular, Bridgeman et al. explored the MBOs in inorganic systems, showing how its basis-set dependence leads to variations in predicted bond orders for transition-metal chloride complexes and sulfur-nitrogen rings (Bridgeman et al., 2001).

Within AIM theory, the electronic density is partitioned into atomic basins, and bond critical points (BCPs) are identified along the bond paths. Bond orders derived from AIM definitions depend sensitively on the chosen atomic partitioning scheme. For instance, Mulliken and Löwdin approaches can lead to substantial basis-set dependence. In contrast, Natural Atomic Orbitals (NAOs) and Intrinsic Atomic Orbitals (IAOs) provide orthonormalized and minimally biased representations of atomic contributions, offering a more stable and chemically intuitive decomposition (Knizia, 2013).

The Atoms in Molecules AIM framework provides a broad conceptual and computational platform for analyzing chemical bonding by partitioning a molecular system into well-defined atomic contributions. Although AIM is often associated with the Quantum Theory of Atoms in Molecules (QTAIM), which is based on real-space partitioning of the electron density using zero-flux surfaces, the AIM concept is not limited to this single approach. In fact, several alternative real-space partitioning schemes exist, such as Hirshfeld, Becke, Iterative Hirshfeld (Hirshfeld-I), and related methods, each differing in how atomic basins are defined and how electron density is assigned to individual atoms. These approaches offer complementary perspectives and may be better suited for specific chemical problems, particularly when dealing with charge

transfer, polarization, or basis-set dependence.

Beyond real-space descriptions, the AIM framework also encompasses Hilbert-space, or orbital-based, definitions of atoms and bonding. In this context, chemical information is extracted from molecular orbitals, density matrices, or localized orbital representations rather than directly from the electron density in real space. Examples include population analyses based on atomic orbitals, natural bond orbital (NBO) analysis, and energy decomposition schemes. Together, real-space and Hilbert-space formulations highlight the conceptual richness of AIM, demonstrating that atomic partitioning is not unique but method-dependent, and that different definitions can provide complementary insights into electronic structure, bonding, and reactivity.

The QTAIM formalizes bonding by partitioning real space into non-overlapping atomic basins defined by the topology of the electron density. Bonding interactions are then analyzed through properties evaluated within these atomic basins and at critical points of the electron density, allowing both covalent and noncovalent interactions to be characterized (Foroutan-Nejad et al., 2014).

These different definitions can yield quantitatively distinct bond order values, even for the same chemical bond. For example, in a series of transition-metal complexes, the MBO may indicate a nearly double-bond character, whereas QTAIM bond orders often provide lower values due to the explicit consideration of electron density distribution within atomic basins. Similarly, in conjugated organic systems, NAO- and IAO-based bond orders capture subtle electron delocalization effects more consistently than traditional Mulliken populations. By comparing these methods across selected systems, such as metal-ligand bonds in palladium and rhodium complexes or C-C bonds in π -conjugated frameworks, we illustrate that differences in bond order values are not arbitrary but rather reflect the underlying physical partitioning and electron delocalization captured by each definition.

Importantly, incorporating these AIM-based and orbital-based bond orders allows chemists to rationalize mechanistic trends more reliably. For instance, when analyzing oxidative addition or reductive elimination steps in transition-metal catalysis, QTAIM and IAO bond orders can reveal bond weakening or strengthening trends that might correlate with activation energies and reaction selectivity. Such an approach complements the MBO framework, providing a multidimensional view of bonding changes along the reaction coordinate. Recent benchmark studies, such as those using the Multiwfn program, allow direct comparison of Mayer, Wiberg, NAO, and QTAIM bond orders, highlighting divergences particularly in delocalized and multi-center bonding systems (Lu, 2024).

Overall, aside from the well-known deficiencies of Mulliken analyses, we emphasize that no single definition is universally superior; instead, the choice of AIM or orbital-based bond order should be guided by the chemical context and the specific mechanistic questions of interest. By presenting examples across different methodologies, our discussion now illustrates both the limitations and the strengths of each approach, enabling readers to make informed choices in their mechanistic analyses.

6.2. Natural bonding orbitals

After achieving a converged density from KS-DFT, this density is utilized to construct the one-particle reduced density matrix (1-RDM). The natural orbitals (NOs) are then defined as the eigenfunctions of the 1-RDM when it acts as an operator (Weinhold and Landis, 2012):

$$\gamma_1 \Theta_k = p_k \Theta_k (k = 1, 2, \dots) \quad (30)$$

where Θ_k represents the natural orbitals (NOs) of the wavefunction. p_k denotes the occupation number (population) of the eigenfunction Θ_k for the molecular electron density operator γ_1 of Ψ .

Solving this eigenvalue equation for the 1-RDM provides the NOs Θ_k along with their occupation numbers p_k , which indicate the contribution

of each natural orbital to the total electron density. The sum of all p_k values equals the total number of electrons in the system.

Although Kohn-Sham orbitals do not directly correspond to natural orbitals (NOs), the optimized KS density can be used to construct the one-electron reduced density matrix (1-RDM), from which the NOs are obtained. Because they are derived from the 1-RDM, NOs provide a more chemically meaningful and accurate representation of the electron density, making them particularly valuable for analyzing molecular electronic structure.

To refine their chemical interpretation, the NOs are typically localized at atomic positions, transforming into one-center localized orbitals known as "natural atomic orbitals" (NAOs). Each NAO maintains intraatomic orthogonality with other NAOs on the same atom and interatomic orthogonality with those on different atoms:

$$\langle \Theta_j^{(A)} | \Theta_k^{(B)} \rangle = \delta_{j,k} \delta_{A,B} \quad (31)$$

By removing interatomic orthogonality, "pre-orthogonal NAOs" (PNAOs) are obtained:

$$\langle p\Theta_j^{(A)} | p\Theta_k^{(A)} \rangle = \delta_{j,k} \quad (32)$$

$$\langle p\Theta_j^{(A)} | p\Theta_k^{(B)} \rangle \neq 0 \quad (33)$$

Unlike NAOs, PNAOs can overlap with orbitals on neighboring atoms, making them more similar to an "idealized" depiction of atomic orbitals and better representing bonding interactions within a molecule.

Natural hybrid orbitals (NHOs) are obtained by forming linear combinations of NAOs through an optimization process that minimizes the total energy of the system. This is achieved by solving a secular equation involving the Hamiltonian and the NAO basis:

$$h_A = \sum a_k \Theta_k^{(A)} \quad (34)$$

The nature of the resulting NHOs depends on their composition:

A single normalized NHO corresponds to a one-center orbital, typically describing lone pairs (LPs):

$$n_A = h_A \quad (35)$$

A combination of two natural hybrid orbitals (NHOs) forms a two-center bonding orbital, corresponding to a NBO. This localization scheme is based on an underlying Lewis structure, which constitutes one of its main limitations:

$$\Omega_{AB} = a_A h_A + a_B h_B; \quad a_A^2 + a_B^2 = 1 \quad (36)$$

The normalized polarization coefficients provide insight into the bonding nature:

→ $a_A = a_B$ indicates a covalent bond.

→ $|a_A - a_B| \approx 1$ suggests an ionic bond.

Each bonding interaction Ω_{AB} also has a corresponding out-of-phase interaction:

$$\Omega_{AB}^* = a_A h_A - a_B h_B \quad (37)$$

These out-of-phase orbitals, classified as non-Lewis orbitals in the NBO6 framework (Glendening et al., 2013), play a crucial role in resonance stabilization and delocalization effects.

6.3. Effective atomic orbitals leading to oxidation states

The Effective Oxidation State (EOS) method, developed by Salvador and co-workers, provides a general and robust theoretical framework for assigning oxidation states directly from the electronic wavefunction (Salvador et al., 2024). In detail, the EOS method provides a consistent way to quantify oxidation states based on electron distribution derived from the wavefunction itself, offering a unified description of electron transfer and redox behavior across diverse chemical systems. EOS has

been successfully applied to molecular complexes, transition-metal compounds (Monreal-Corona et al., 2025), organometallic systems (Joly et al., 2023), clusters, and solids (Comas-Vilà et al., 2025), allowing for an unbiased description of electron distribution, metal–ligand bonding, redox processes, and charge transfer phenomena. In particular, it has proven valuable for analyzing ambiguous or non-classical oxidation states, metal–metal bonding, and delocalized electronic structures. However, despite its broad applicability, EOS analyses within DFT remain partially limited by the intrinsic single-determinant nature of most functionals and their treatment of static electron correlation. In addition, Salvador, Gimferrer, and co-workers have recently advanced this theoretical foundation by introducing a wavefunction-based localization framework that allows for an even more precise and transferable definition of atomic contributions in molecules (Salvador et al., 2024). Their approach extracts effective atomic orbitals (EffAOs) through independent localization transformations for each atom (Mayer and Salvador, 2009; Ramos-Cordoba et al., 2013; Gimferrer et al., 2020), derived directly from the molecular wavefunction, enabling an accurate depiction of atomic or fragment electron configurations. This methodology refines the concept of oxidation states by linking them explicitly to the localized electronic structure, offering a nuanced understanding of electron distribution and chemical bonding. The resulting framework provides a consistent and quantitative way to interpret charge transfer, redox activity, and bond polarity in complex systems, and it has been validated across a broad range of challenging molecular cases, further strengthening the applicability of EOS and its extensions to real-world catalytic processes.

For systems described by single-determinant wavefunctions, we have also included references to complementary methods that provide oxidation-state analysis through localized orbitals, such as the Localized Orbital Bonding Analysis (LOBA) developed by Head-Gordon and co-workers (Thom et al., 2009), its Intrinsic Bond Orbital (IBO) extension proposed by the Klein group (Leach and Klein, 2024), and the Oxidation State Localized Orbitals (OSLO) approach developed jointly by Salvador and Head-Gordon (Gimferrer et al., 2022).

6.4. Energy decomposition analysis

The Energy Decomposition Analysis (EDA) has become an indispensable tool for elucidating the physical factors governing chemical bonding and reactivity (Ziegler and Rauk, 1977; Morokuma, 2025). The method partitions the interaction energy between molecular fragments into physically interpretable components, typically electrostatic, Pauli repulsion, orbital interaction, and dispersion terms, thereby providing a quantitative framework to dissect the nature of a chemical bond. The classical EDA scheme introduced by Ziegler and Rauk within the context of the Kohn–Sham molecular orbital framework allows one to distinguish between the steric and electronic contributions to bonding (Ziegler and Rauk, 1979), with a broad range of applications in the characterization of ligands of complexes (Poater et al., 2006; Bantreil et al., 2012; Poater et al., 2010). More recently, this approach has been extended by Bickelhaupt and Houk to study reaction mechanisms through the Activation Strain Model (ASM), which analyzes the evolution of both the intrinsic strain of the reactants and their mutual interaction along the reaction coordinate (Fernández and Bickelhaupt, 2014). This combined EDA–ASM framework has proven highly insightful for rationalizing reactivity trends, selectivity, and activation barriers in a wide range of organic, organometallic, and catalytic transformations (Bickelhaupt and Houk, 2017; Vermeeren et al., 2021).

In parallel, alternative decomposition methods have emerged, such as the Interacting Quantum Atoms (IQA) approach (Casals-Sainz et al., 2020), which is grounded in the Quantum Theory of Atoms in Molecules (QTAIM). IQA partitions the total energy into intra- and interatomic contributions based on real-space integration, offering a complementary, density-based perspective on chemical bonding (Tomasini et al., 2026). This methodology has been successfully applied to study covalent

and noncovalent interactions, reaction mechanisms, and cooperative effects in complex systems.

Actually, it should be emphasized that there is no unique or exact EDA scheme, and consequently no universal set of energy components, as Andrés et al. announced (Andrés et al., 2019). For instance, in ZR-EDA the interaction energy is partitioned into Coulomb, Pauli repulsion, dispersion, and orbital interaction terms, whereas alternative frameworks such as ALMO-EDA or SAPT rely on different conceptual and mathematical decompositions. Furthermore, the IQA approach does not need to be strictly grounded in QTAIM; other real-space partitioning schemes, and even Hilbert-space formulations, have been reported, although the latter often lose a direct connection to chemically intuitive energy scales. Finally, while most EDA schemes are traditionally applied to single-determinant wavefunctions, there exist well-established extensions to correlated methods, including ALMO-EDA-MP2, Local Energy Decomposition (LED), and IQA applied to correlated wavefunctions, underscoring the broader applicability of energy decomposition concepts beyond the mean-field level.

6.5. NCI plots

The common gradient of electron density serves as an effective tool for identifying strong bonding interactions, much like the Quantum Theory of Atoms in Molecules (QTAIM) and other density-based topological methods. This is because it highlights regions of concentrated electron density (Bader, 1991, 2003; Alonso et al., 2014). However, for non covalent interactions (NCIs) (Bauzá et al., 2015; Rondelli et al., 2024), the critical points in the density are less pronounced. These interactions typically involve subtle variations in electron density, making critical points less distinct and harder to detect using conventional gradient-based approaches (see Fig. 6).

The Non-Covalent Interaction (NCI) index is a visualization tool derived from electron density and its derivatives, enabling the identification of NCIs. At critical points between interacting molecules, the reduced density gradient (s) exhibits significant changes due to the annihilation of the density gradient (Johnson et al., 2010). When s is plotted against density across a molecule (Boro et al., 2024), sharp peaks indicate the presence of NCIs. The reduced density gradient is given by:

$$s = (1/c_s)(|\nabla\rho|/\rho^{4/3}); \text{ where } c_s = 2(3\pi^2)^{1/3} \quad (38)$$

Furthermore, the nature of NCIs, whether stabilizing or destabilizing, can be assessed by analyzing the eigenvalues of the second derivative (Hessian) of the electron density (Boto et al., 2016). This evaluation

provides deeper insight into the interaction characteristics, and allows the rationalization of NCIs (Bazvand et al., 2023; Dey et al., 2023).

Contreras, Mata, and coworkers have recently extended the analysis of NCIs and dispersion interaction densities (DID) through their decomposition into pairwise orbital contributions, opening new avenues for data-driven and machine-learning methodologies (Aniban et al., 2025).

While the NCI plot remains the most widely used visualization tool (Johnson et al., 2010), it is inherently based on electron density and its derivatives, providing qualitative insight into the spatial distribution and strength of weak interactions. To complement this density-based picture, we now discuss Dispersion Interaction Density (DID) analysis, developed by Mata and coworkers (Wuttke and Mata, 2017). DID enables a quantitative decomposition of dispersion contributions at the correlated wavefunction level, resolving them into pairwise orbital contributions. This method offers a detailed understanding of how dispersion energy is distributed across molecular space and among interacting fragments, thereby enriching the interpretation of NCIs obtained from standard density descriptors.

Additionally, we have included a discussion of the Local Energy Decomposition (LED) approach proposed by Bistoni, Neese, and coworkers (Bistoni, 2020; Bistoni et al., 2024). LED provides a framework for decomposing the interaction energy derived from coupled-cluster wavefunctions into physically meaningful terms such as electrostatics, exchange, polarization, and dispersion. This energetic decomposition complements density-based descriptors like NCI and allows a rigorous energetic quantification of non-covalent effects.

6.6. Machine learning: a tool to use the right descriptor to describe the reactivity

We emphasize that the concepts discussed throughout this review have broad applications in computational catalysis, reaction mechanism analysis, and predictive chemistry. For instance, descriptors derived from DFT calculations, such as frontier molecular orbitals, bond orders, non-covalent interaction indices, and energy decomposition analysis, can be directly integrated with machine learning models to predict reactivity, selectivity, and catalyst performance. Modern ML approaches, including supervised learning, regression models, neural networks, and ensemble methods, have been successfully applied to correlate chemical descriptors with experimental outcomes, enabling accelerated screening of catalysts and reaction conditions. In this context, software tools such as ROBERT provide a versatile platform to

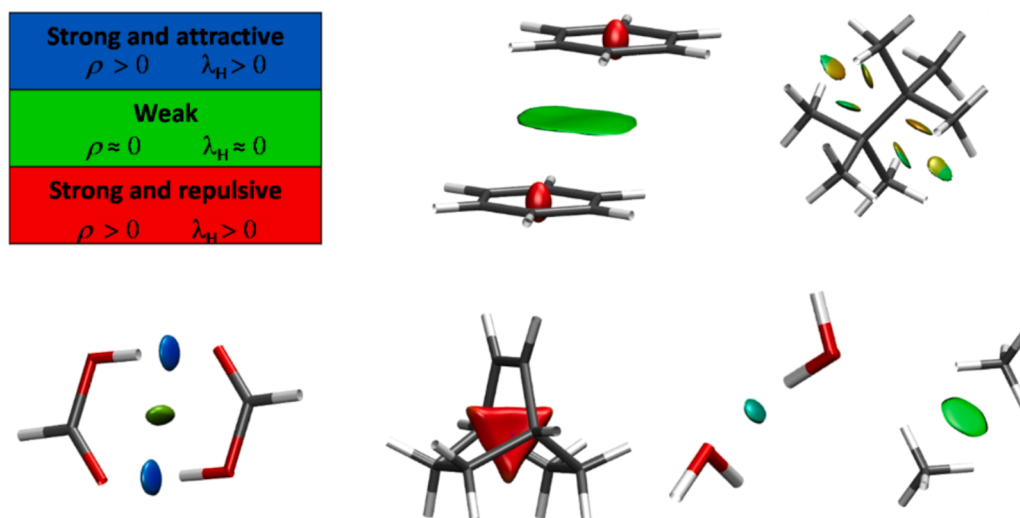


Fig. 6. Image from the official NCIPlot program website, theoretical section. The color of the depicted NCIs is determined by the eigenvalues of the second density Hessian (Contreras-García et al., 2011; Contreras, 2025).

bridge chemical data with ML workflows (Dalmau and Alegre-Requena, 2024), offering functionalities for descriptor calculation, dataset curation, model selection, and feature importance analysis. Other widely used tools include ChemML, DeepChem, and the Open Catalyst Project, which collectively facilitate data-driven discovery by combining quantum chemical calculations with ML predictions. By integrating these approaches, chemists can move beyond manual intuition, leveraging computational and statistical methods to rationally design catalysts, predict reaction outcomes, and optimize reaction conditions, thus illustrating the practical impact of the theoretical concepts presented in this review.

7. Conclusions

Computational methodologies have become indispensable tools across multiple scientific disciplines, significantly enhancing our ability to model, simulate, and analyze complex systems. This review apart from revising the theory behind DFT applications, the foundation for solving chemical problems, has highlighted the critical role of computational techniques in advancing fields such as bioinformatics, materials science, and artificial intelligence. The increasing availability of high-performance computing resources, coupled with the rapid development of machine learning algorithms, has enabled researchers to process vast amounts of data, optimize models, and uncover new scientific insights at an unprecedented pace.

DFT modeling has evolved into a central pillar of modern catalysis research, offering mechanistic insight and predictive power for transition states, reaction networks, and catalyst design. Advances in functionals, automation, and data-driven methods now allow accurate, large-scale exploration of catalytic landscapes that were once computationally prohibitive. The integration of DFT with machine learning and kinetic modeling heralds a new era of computational catalysis, one where predictive simulation drives experimental innovation, accelerating the discovery of efficient, sustainable catalytic systems across chemistry and materials science.

One of the key takeaways from this review is the transformative impact of machine learning and artificial intelligence on computational research. These technologies have facilitated automated data analysis, pattern recognition, and predictive modeling, significantly reducing the time and effort required for complex computations. Additionally, emerging fields such as quantum computing and deep learning offer exciting prospects for solving problems that are currently intractable with classical computational approaches. Despite these advancements, several challenges remain, including computational cost, data quality, and the interpretability of machine learning models. Addressing these issues will require continued methodological improvements and interdisciplinary collaboration.

Furthermore, the integration of computational methods with experimental techniques has proven to be highly beneficial. Computational models can provide predictive insights that guide experimental research, while experimental validation enhances the reliability of computational findings. This synergy has been particularly impactful in areas such as drug discovery, where computational screening of molecular interactions can significantly accelerate the identification of potential drug candidates.

The future of computational research will depend on the continued development of more efficient algorithms, scalable computing infrastructures, and improved data management strategies (Teale et al., 2022). Advances in cloud computing, parallel processing, and hybrid computational frameworks will further enhance our ability to tackle increasingly complex problems. Moreover, ethical considerations, data privacy, and reproducibility in computational research must also be addressed to ensure responsible and transparent scientific practices.

In conclusion, computational methods have reshaped the landscape of modern research, offering powerful tools for solving scientific and engineering challenges. While limitations persist, the rapid evolution of

computational techniques, combined with interdisciplinary efforts, promises to drive further breakthroughs in the coming years. By embracing new technologies, refining existing methodologies, and fostering collaboration across disciplines, computational science will continue to push the boundaries of knowledge and innovation.

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No new data were created.

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Artur Brotons Rufes: Writing – original draft. **Sergio Posada Pérez:** Writing – review & editing, Supervision, Conceptualization. **Albert Poater:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests.

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